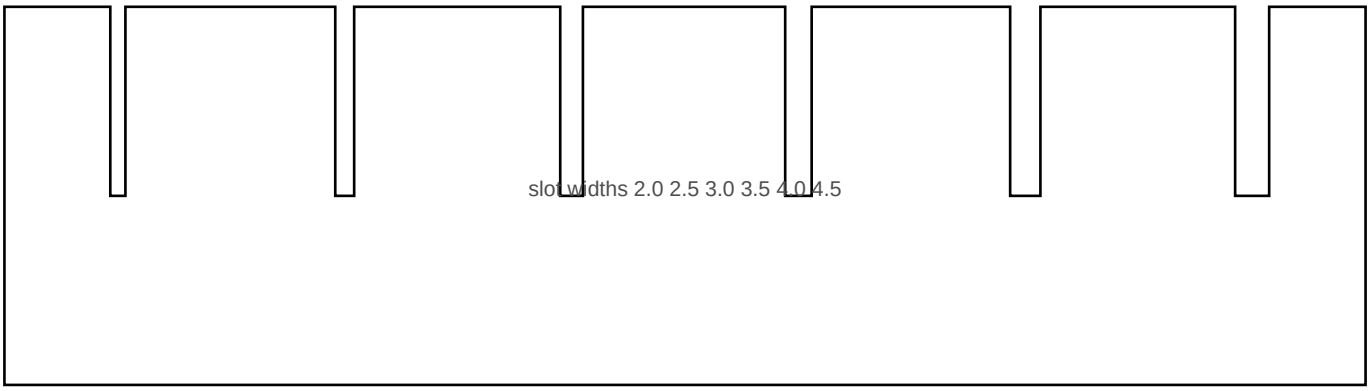
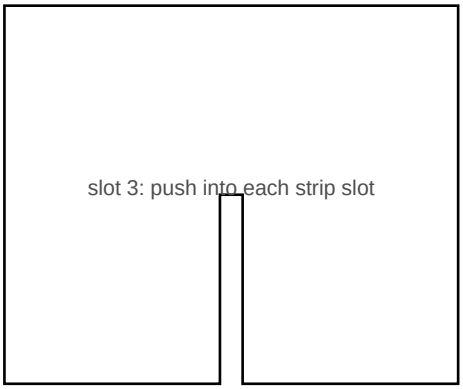
01 Test strip: find your slot width

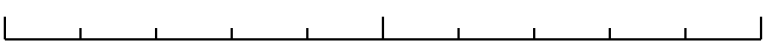
Fits on one A4 sheet. Cut both parts. Push the key into each slot of the strip. The right slot is the narrowest one that goes in fully without tearing. If it is not 3.0, ch

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)





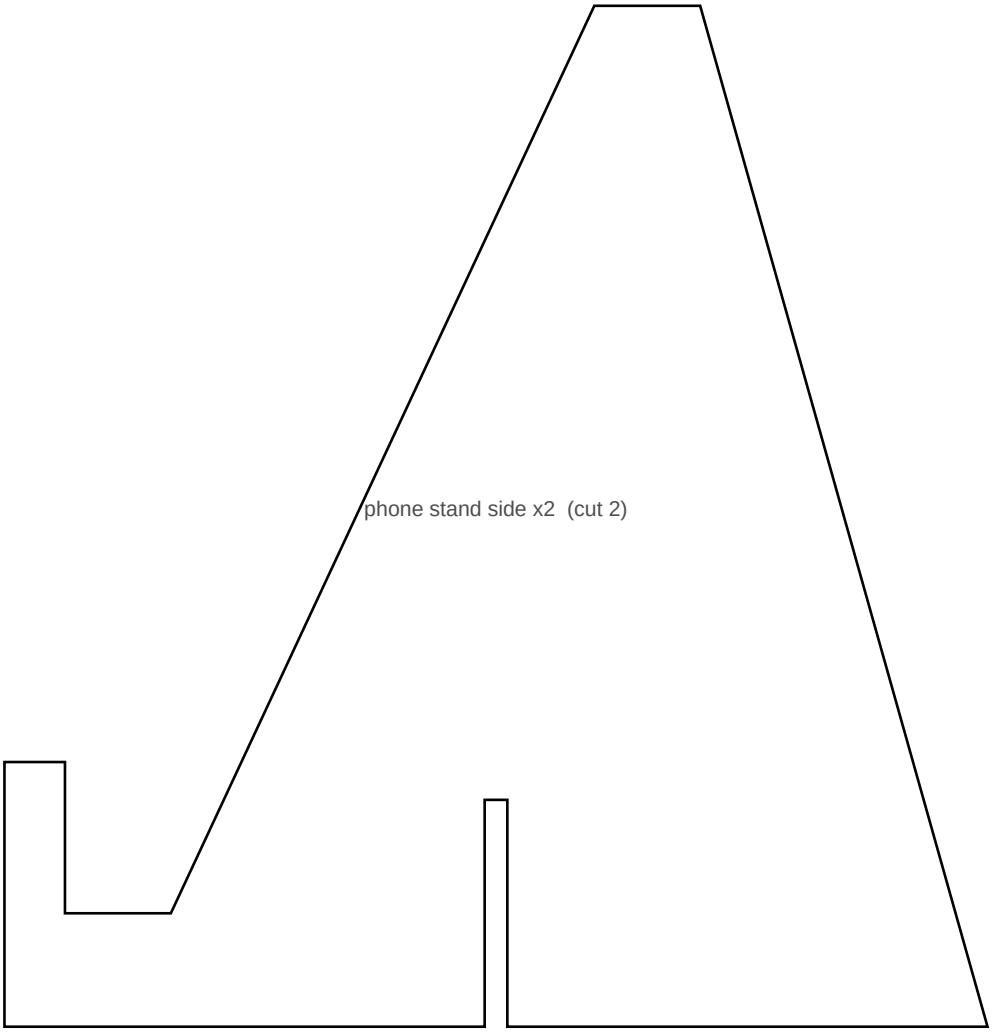
Cut both parts. Push the key into each slot of the strip.
The right slot is the narrowest one that goes in fully without tearing.
If it is not 3.0, change T in generate.py and run it again.

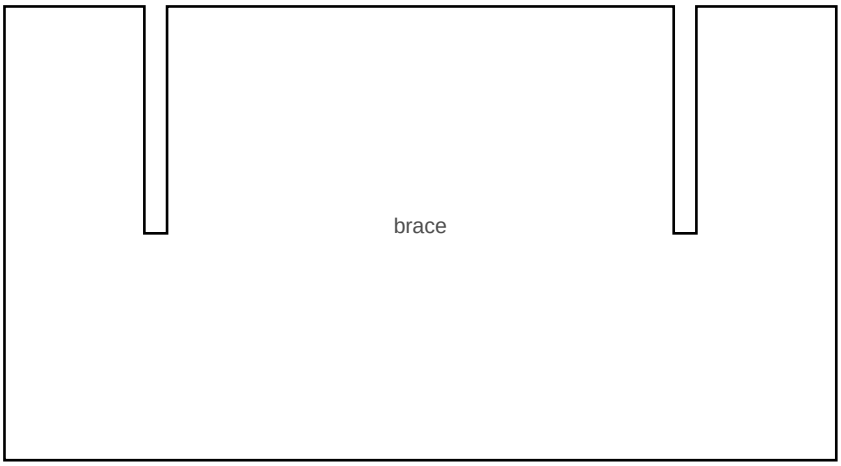


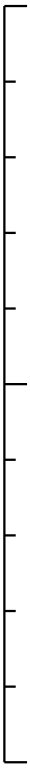
02 Phone stand

Fits on one A4 sheet. Cut 2 sides and 1 brace. Flutes (the waves inside the board) running up-down. Slide the brace slots down onto the side slots. The phone sits on the

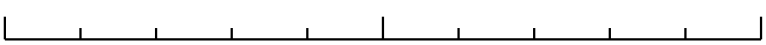
Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)







Cut 2 sides and 1 brace. Flutes (the waves inside the board) running up-down.
Slide the brace slots down onto the side slots. The phone sits on the 14 mm ledge.
Phone up to 12 mm thick with a case; for thicker, move the front lip forward.

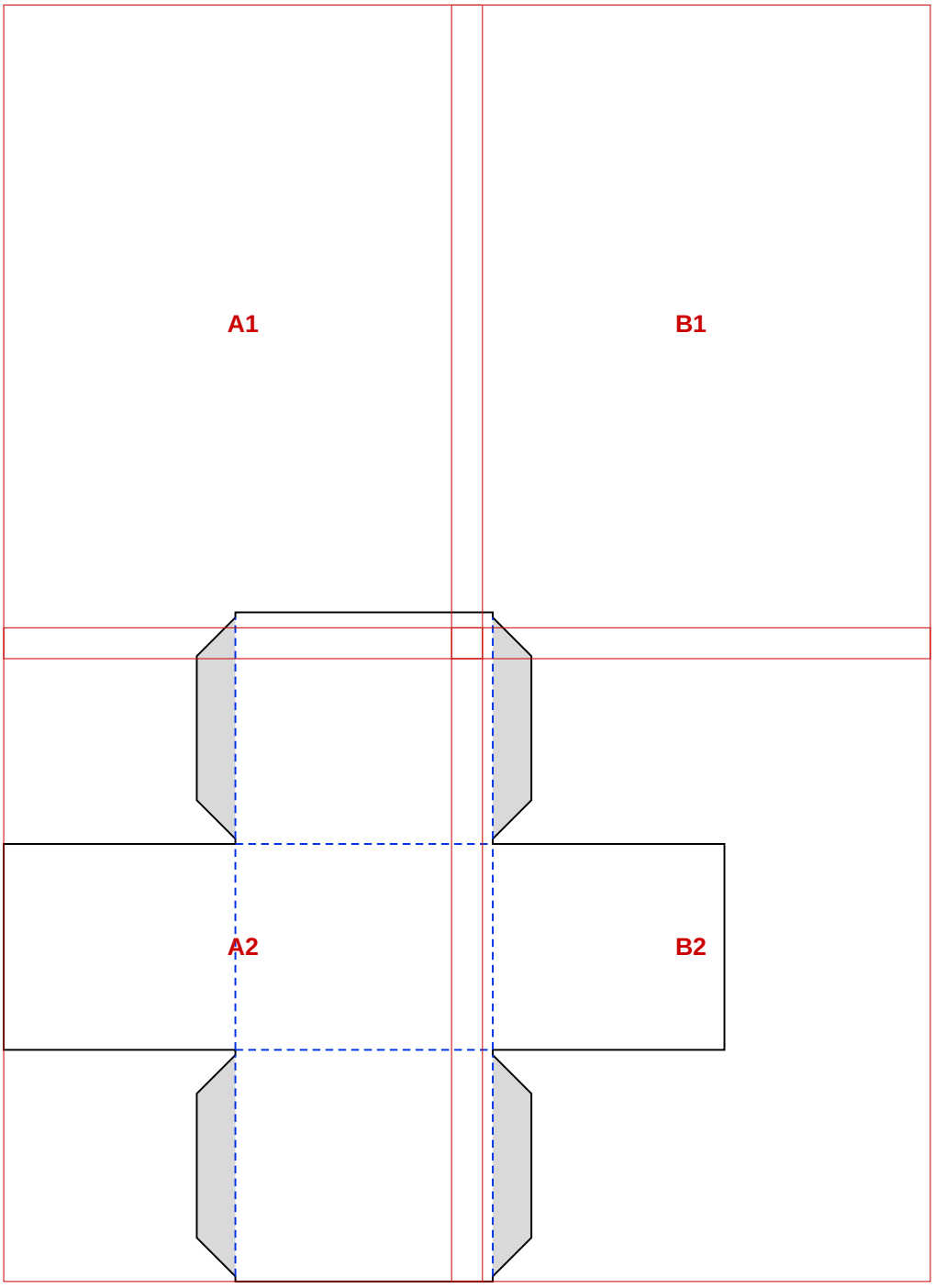


100 mm: measure it! (print at 100 %)

03 Desk organizer box (100 x 80 x 90)

4 A4 sheets (2 across x 2 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: desk organizer, one piece

Tape the sheets together, then cut the outline (solid lines).

Score all dashed lines: press along a ruler with a blunt knife back or an empty pen.

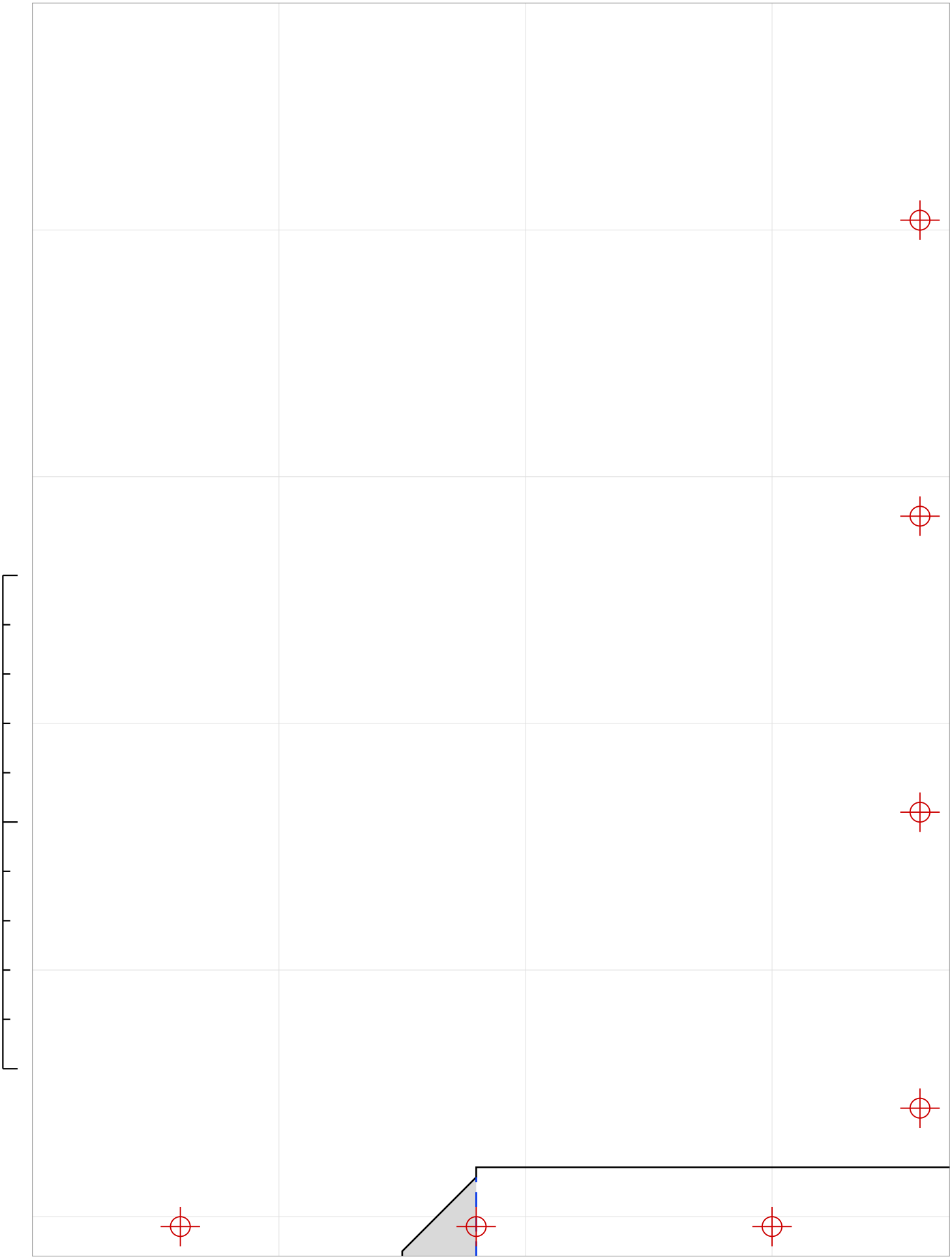
Fold the four walls up. Fold the gray flaps inward and glue them inside the side walls.

100 mm: measure it! (print at 100 %)

03 Desk organizer box (100 x 80 x 90) - sheet A1 (2 x 2)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



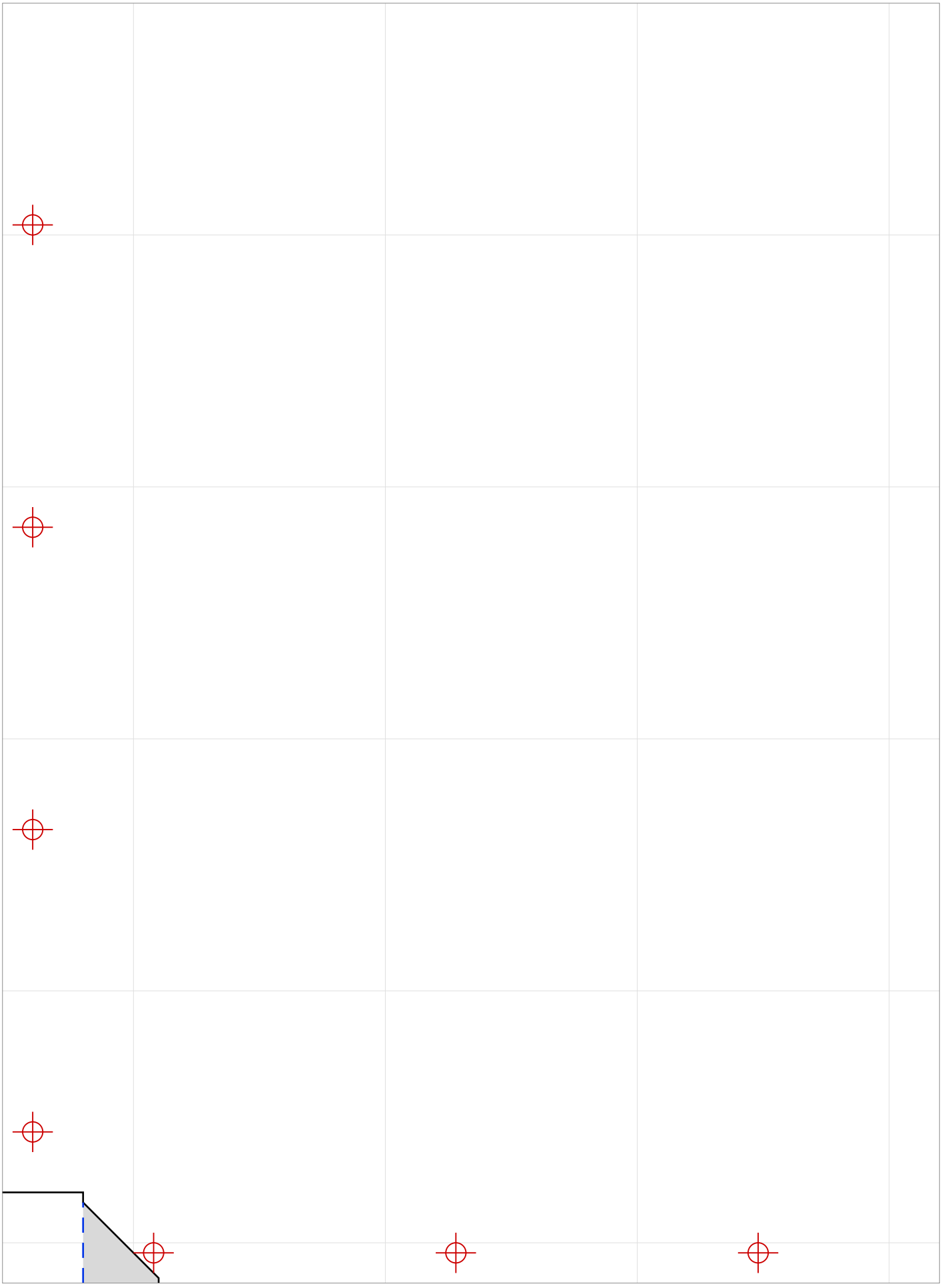
bottom edge overlaps sheet A2

100 mm: measure it! (print at 100 %)

03 Desk organizer box (100 x 80 x 90) - sheet B1 (2 x 2)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



<- left edge overlaps sheet A1

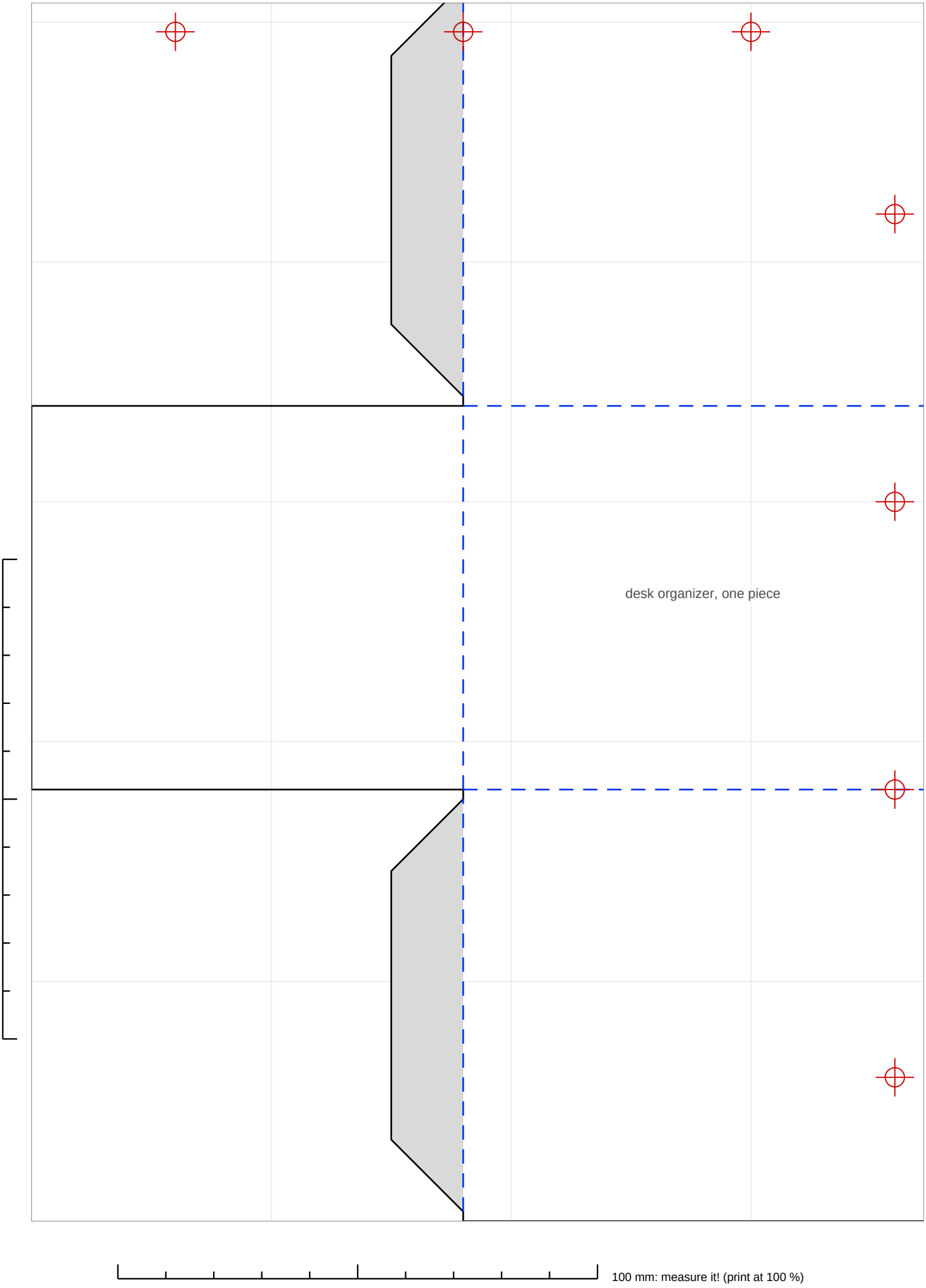
bottom edge overlaps sheet B2

100 mm: measure it! (print at 100 %)

03 Desk organizer box (100 x 80 x 90) - sheet A2 (2 x 2)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

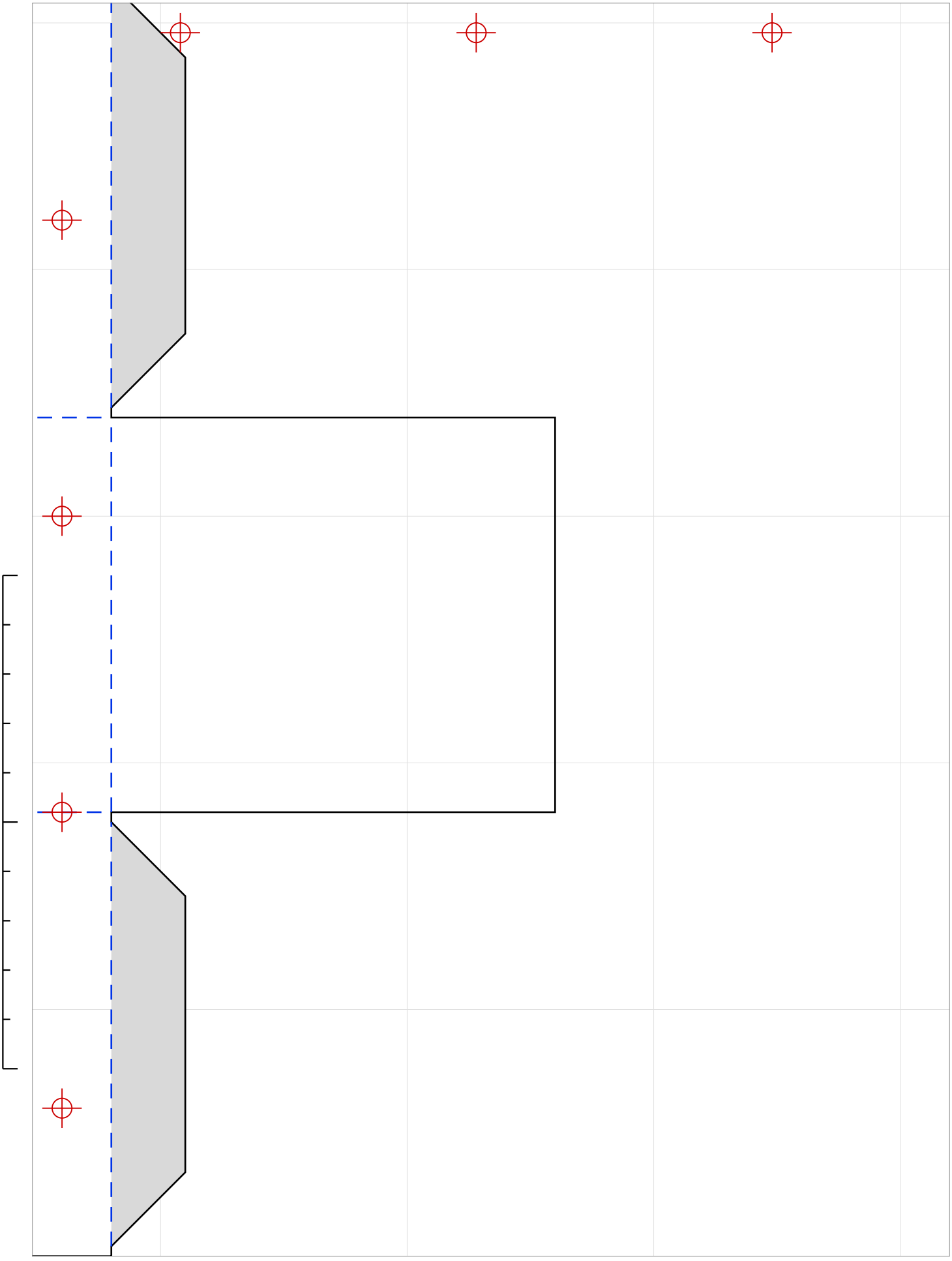
Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



03 Desk organizer box (100 x 80 x 90) - sheet B2 (2 x 2)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



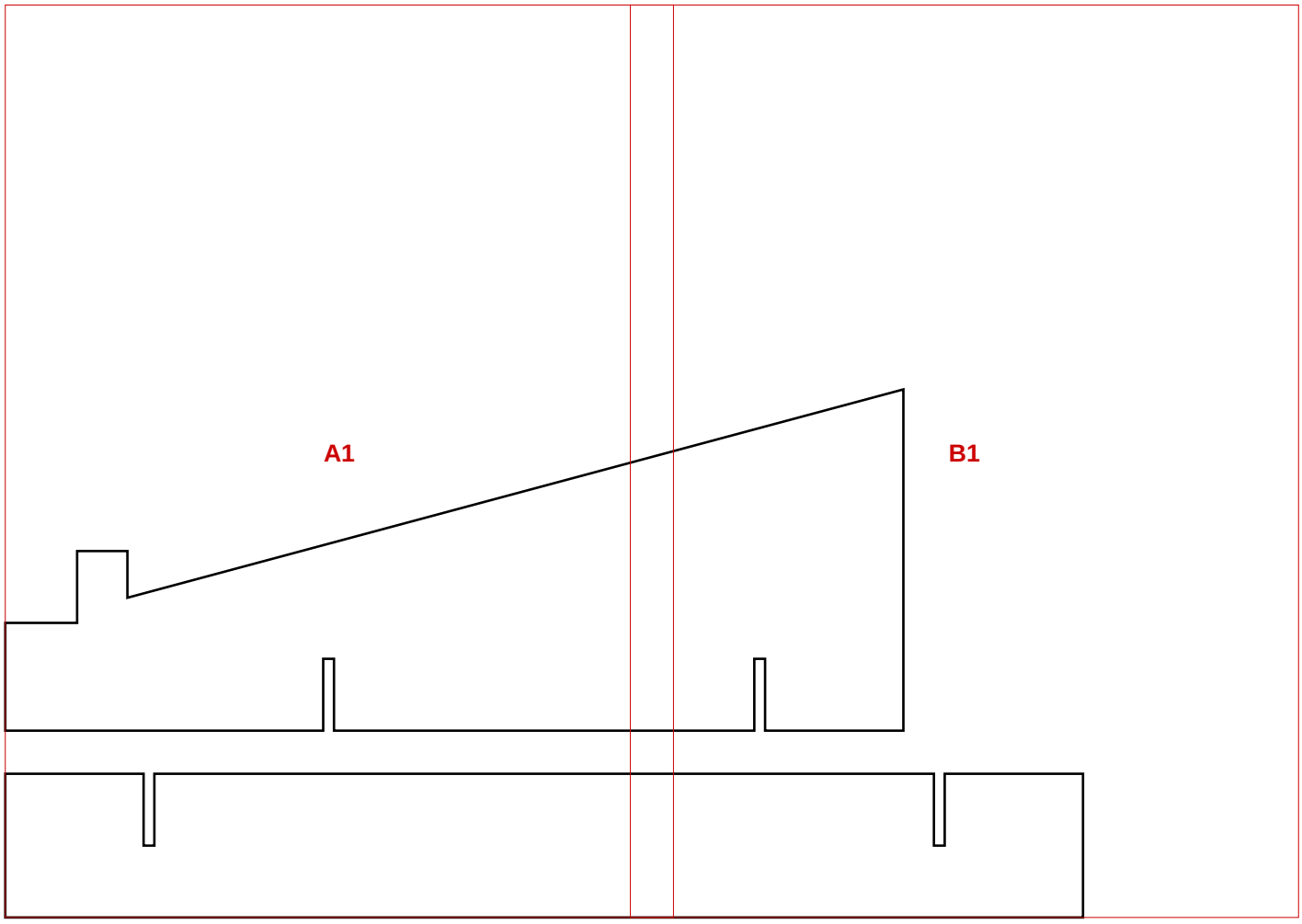
<- left edge overlaps sheet A2

100 mm: measure it! (print at 100 %)

04 Laptop stand

2 A4 sheets (2 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

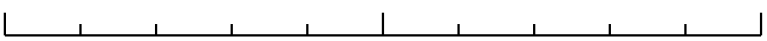
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: laptop stand side x2 (x2), crossbar x2 (x2)

Cut 2 panels and 2 crossbars. Use double-wall board if you have it.

Slide the crossbars into the panel slots (crossbar slots point up).

The lip at the front stops the laptop sliding off. Panels end up 220 mm apart.

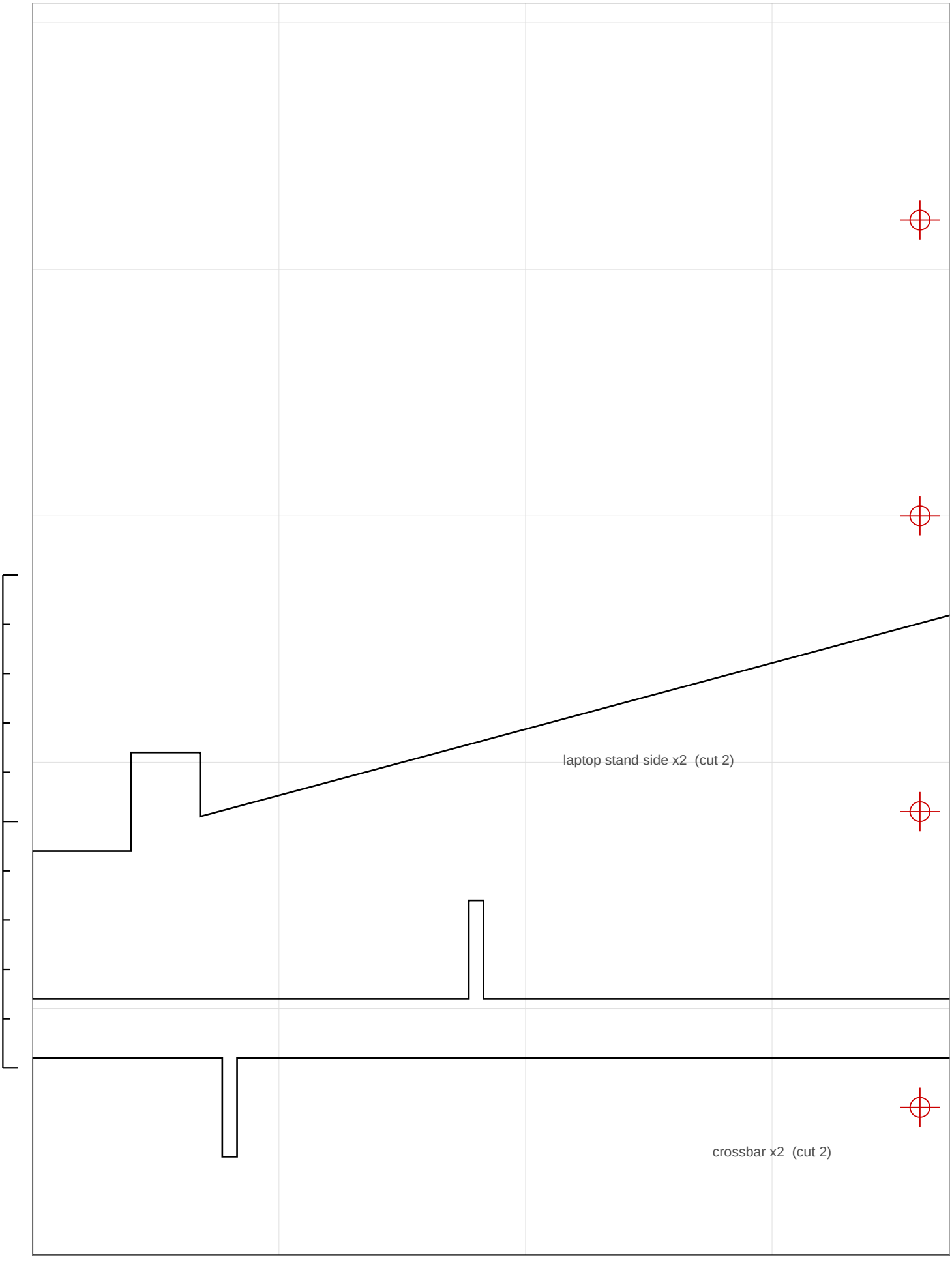


100 mm: measure it! (print at 100 %)

04 Laptop stand - sheet A1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

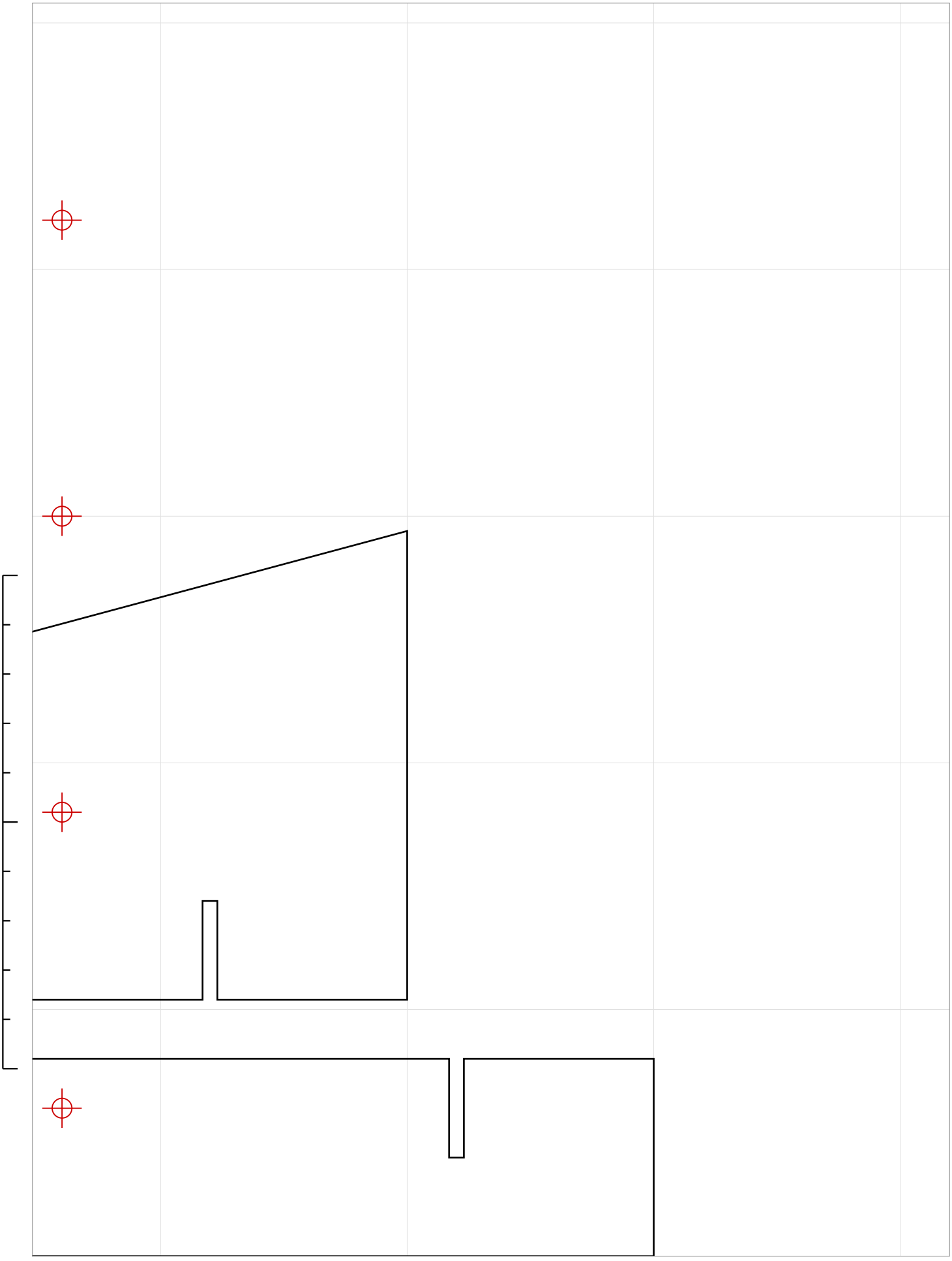


100 mm: measure it! (print at 100 %)

04 Laptop stand - sheet B1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



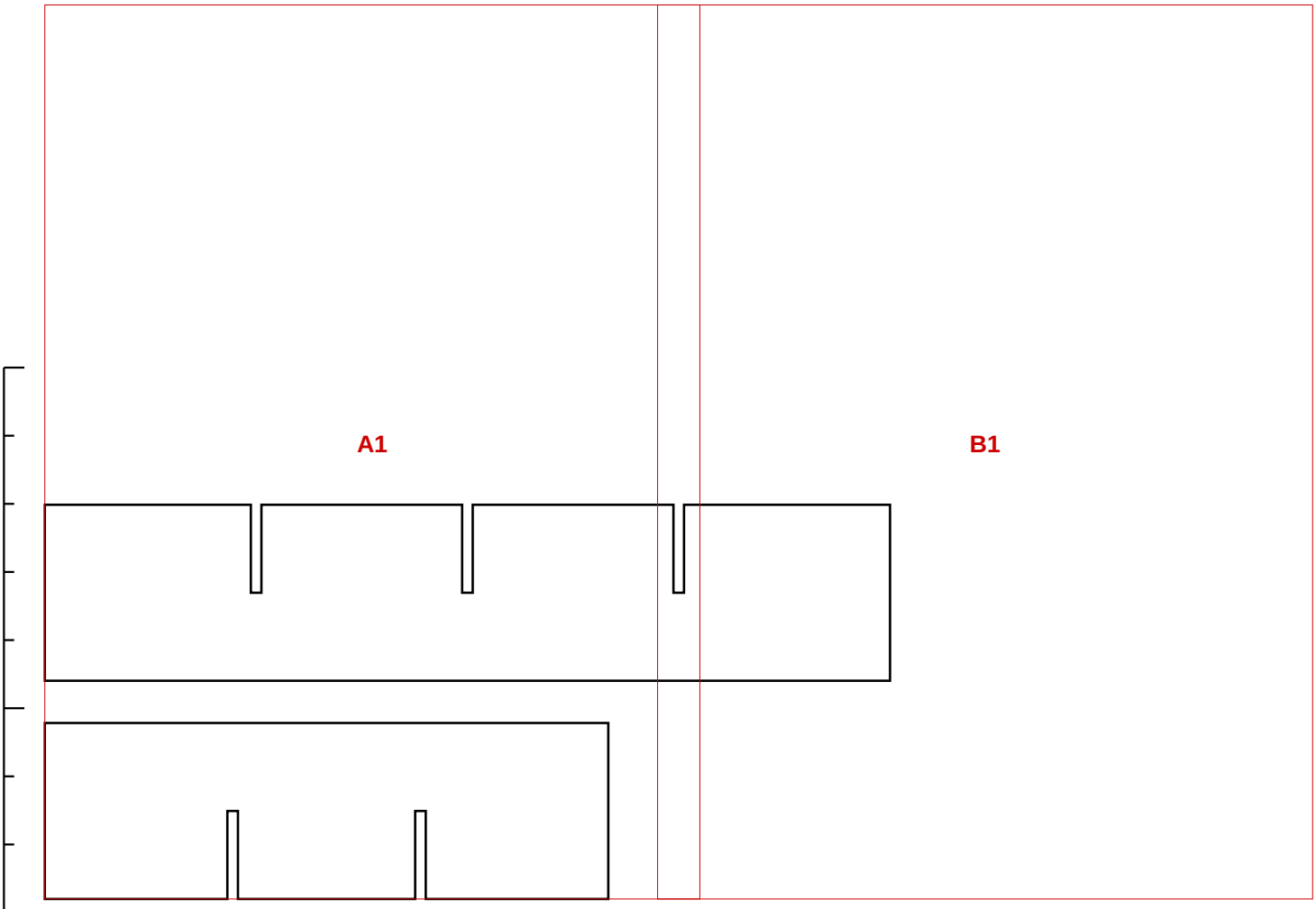
<- left edge overlaps sheet A1

100 mm: measure it! (print at 100 %)

05 Drawer divider grid (240 x 160 x 50)

2 A4 sheets (2 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

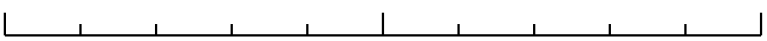
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: long strip x2 (x2), short strip x3 (x3)

Cut 2 long and 3 short strips (flutes running up-down for stiffness).

Slot the short strips down onto the long strips: 12 compartments.

Change L, S, H in generate.py to fit your drawer.

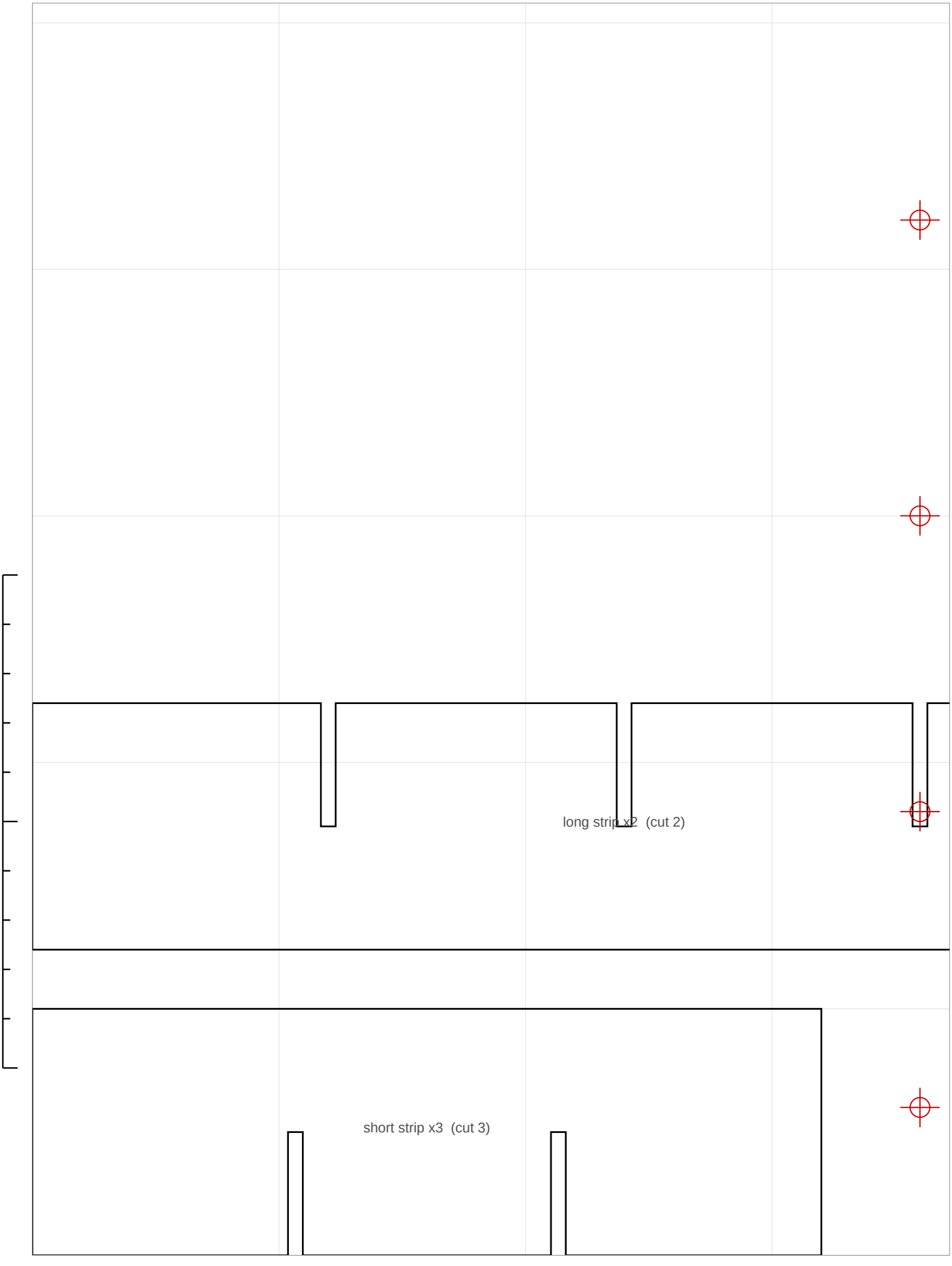


100 mm: measure it! (print at 100 %)

05 Drawer divider grid (240 x 160 x 50) - sheet A1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

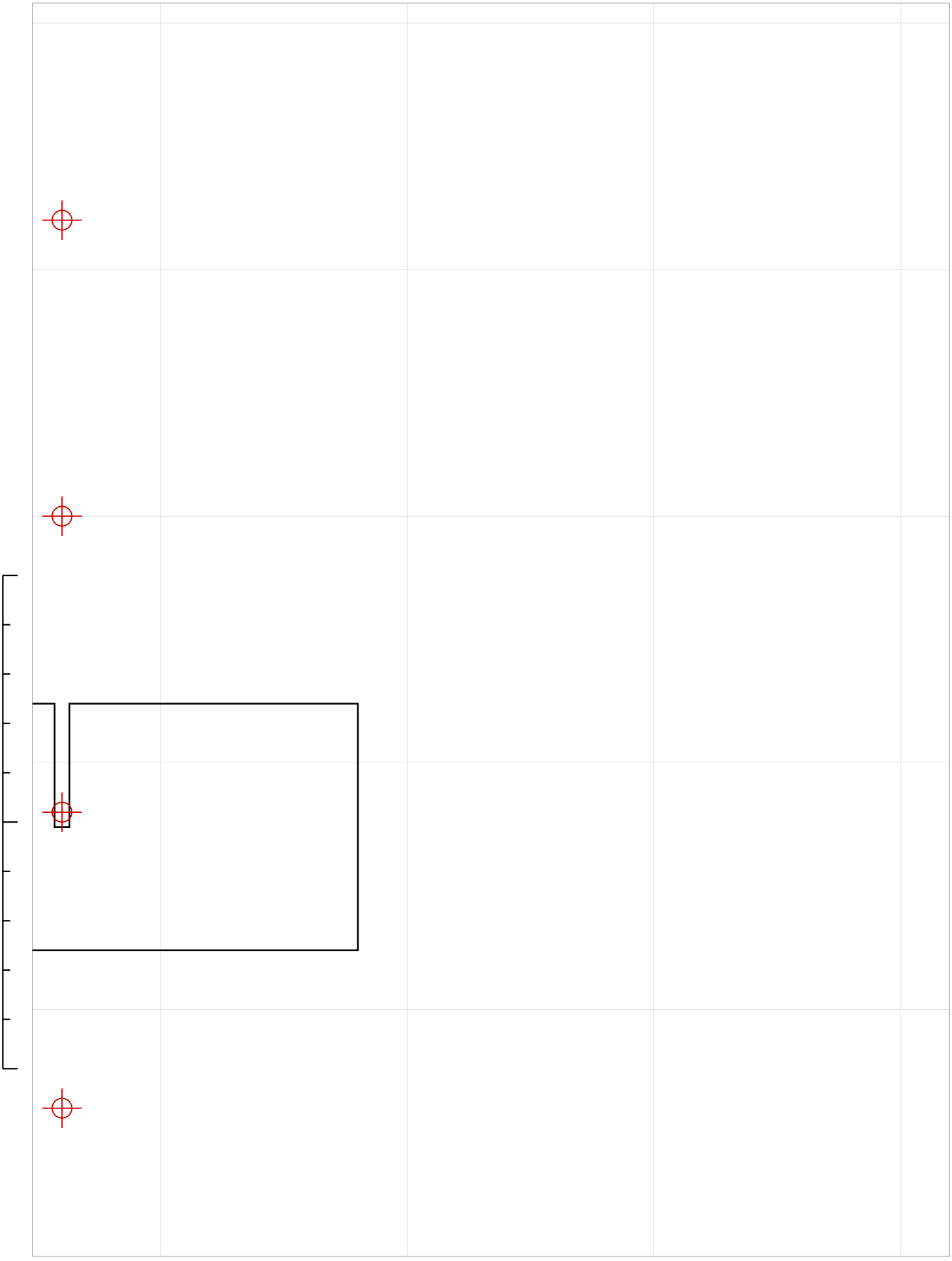


100 mm: measure it! (print at 100 %)

05 Drawer divider grid (240 x 160 x 50) - sheet B1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



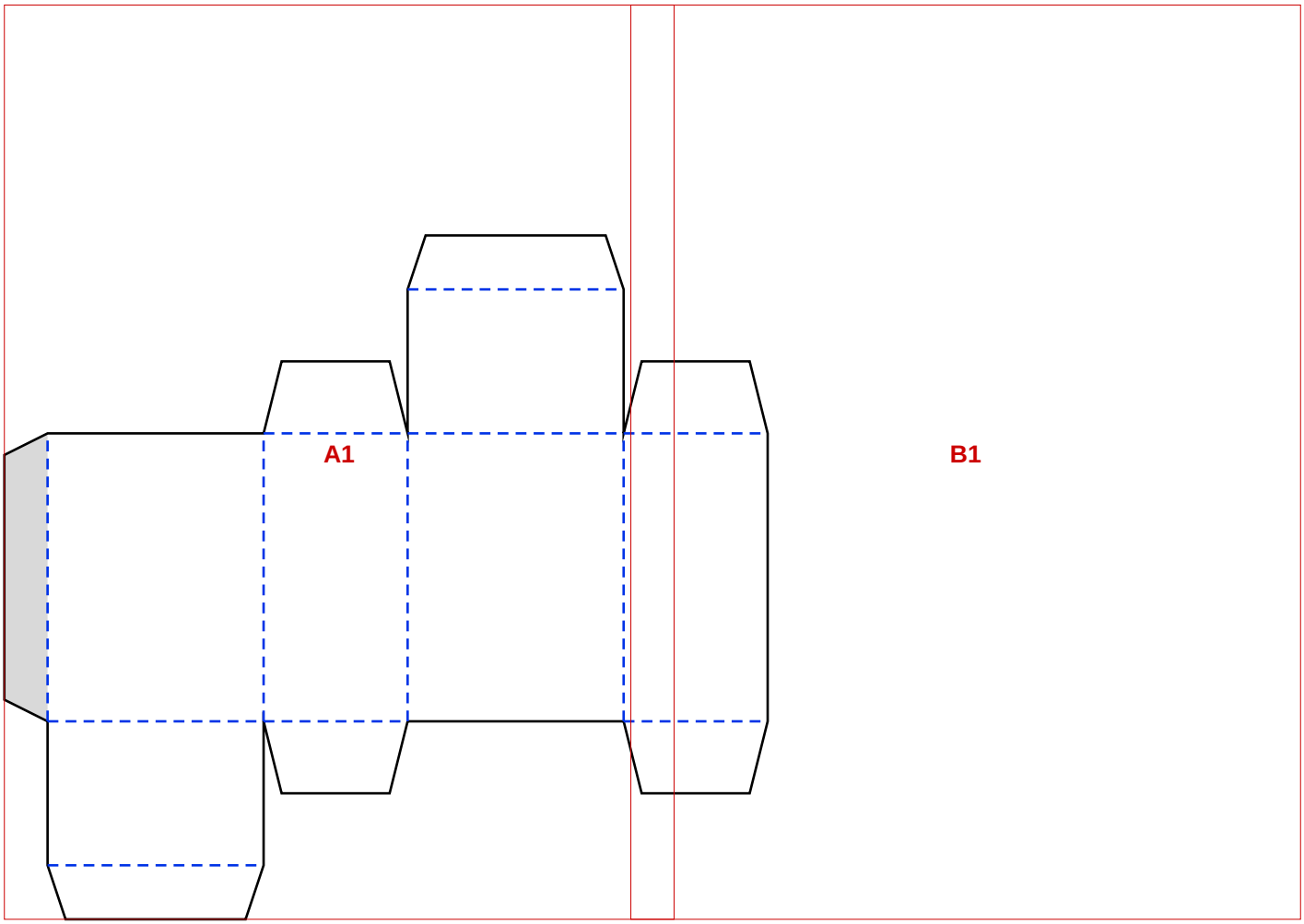
<- left edge overlaps sheet A1

100 mm: measure it! (print at 100 %)

06 Tuck-end box (60 x 40 x 80)

2 A4 sheets (2 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

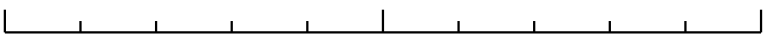
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: tuck box, one piece of thin card

Use thin card (cereal box, 0.3-0.5 mm), not corrugated board.

Cut the outline, score all dashed lines, pre-fold every score.

Glue the gray flap inside the last side panel. Tuck the lids in at both ends.

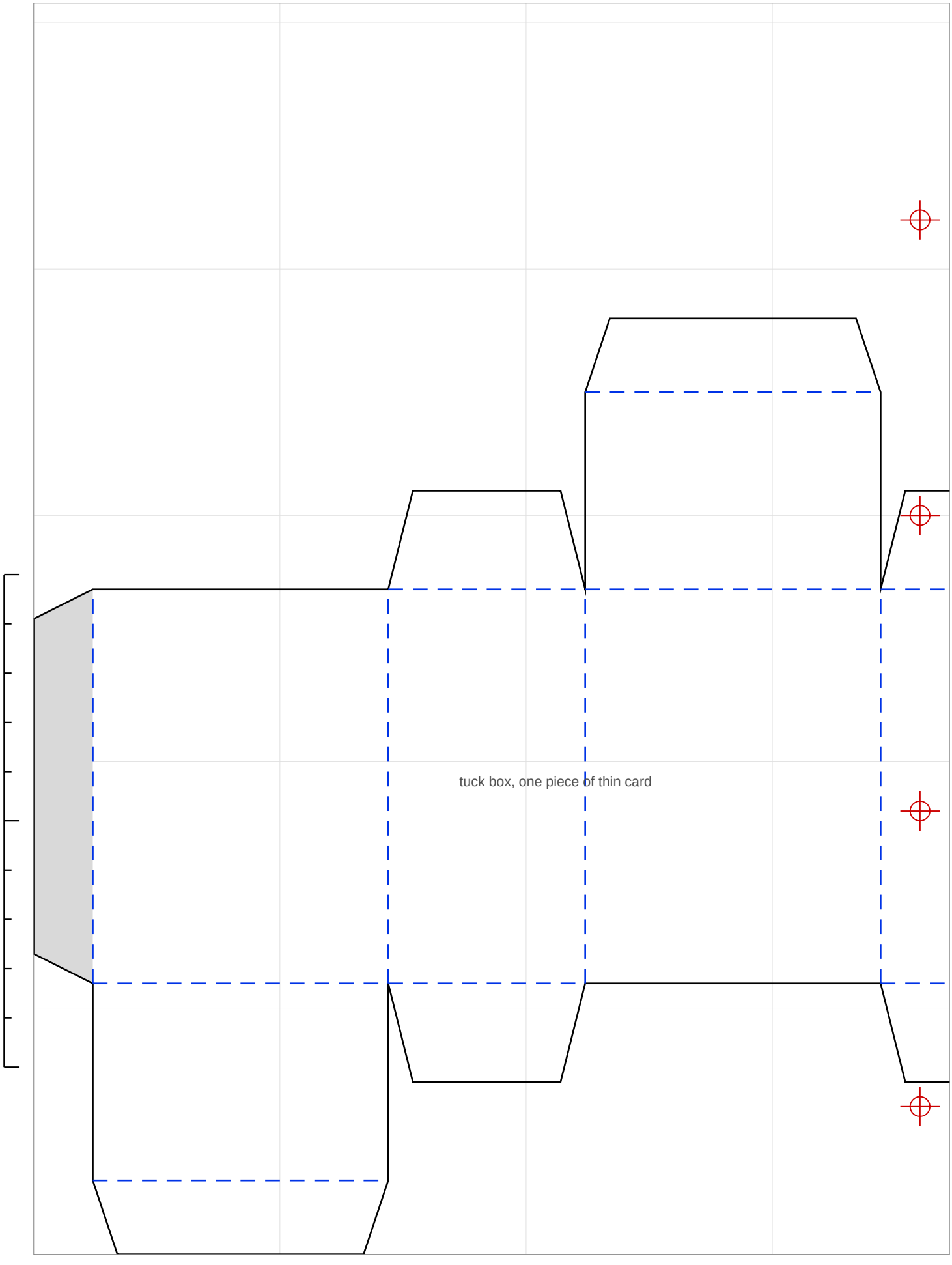


100 mm: measure it! (print at 100 %)

06 Tuck-end box (60 x 40 x 80) - sheet A1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

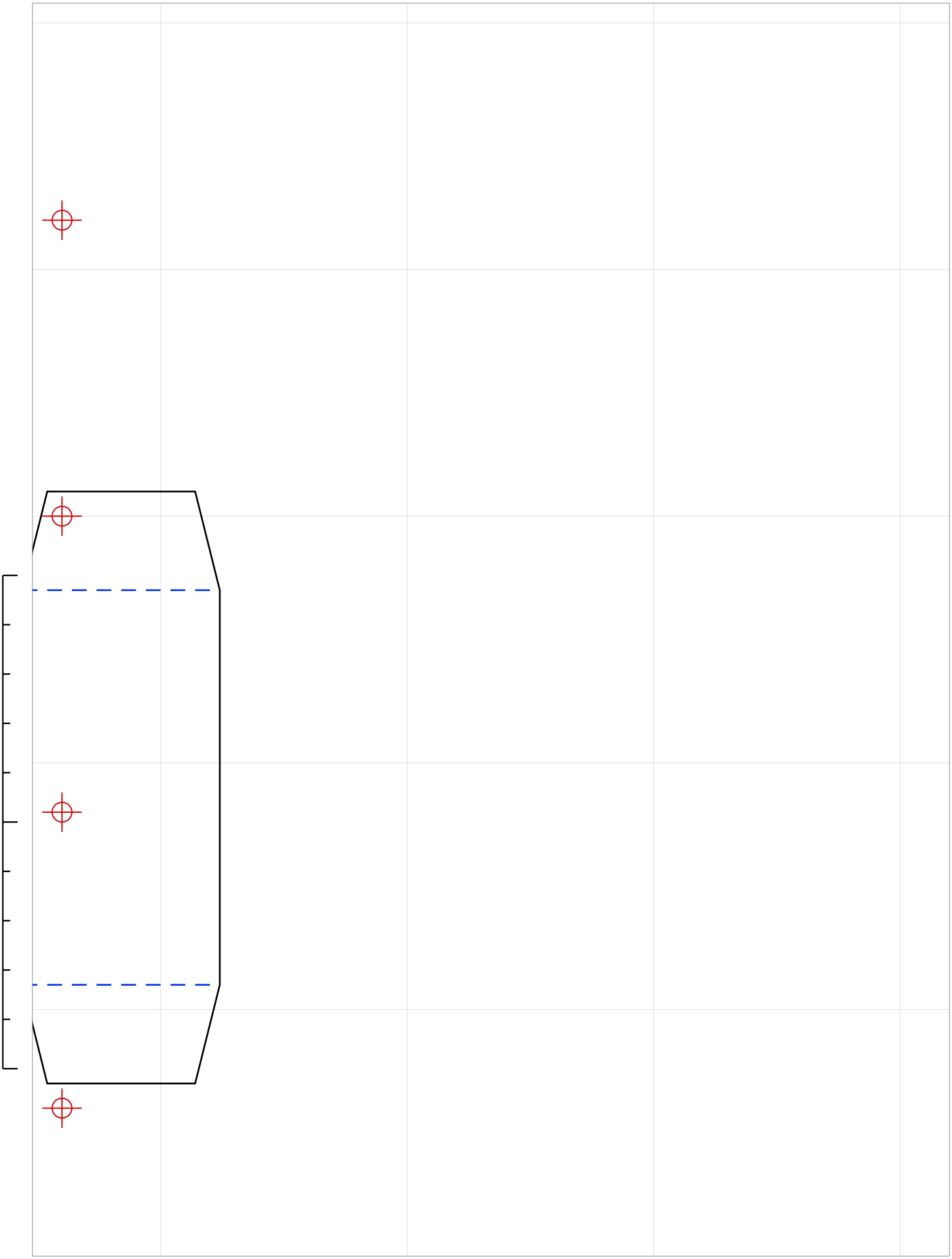


100 mm: measure it! (print at 100 %)

06 Tuck-end box (60 x 40 x 80) - sheet B1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



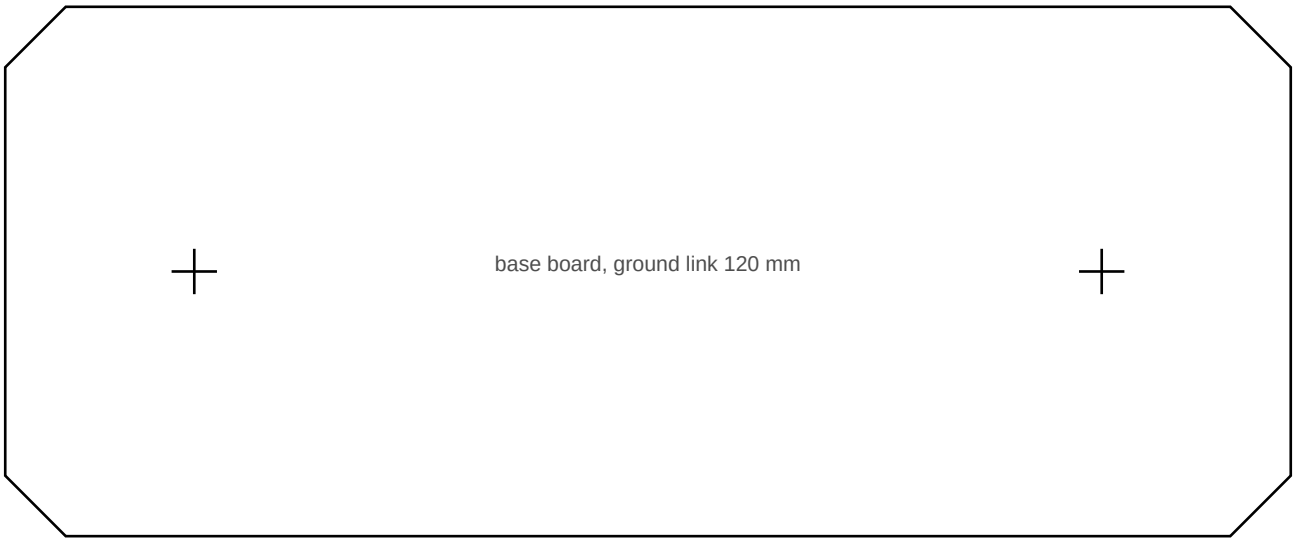
<- left edge overlaps sheet A1

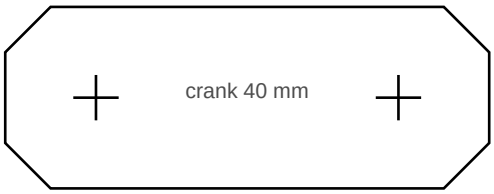
100 mm: measure it! (print at 100 %)

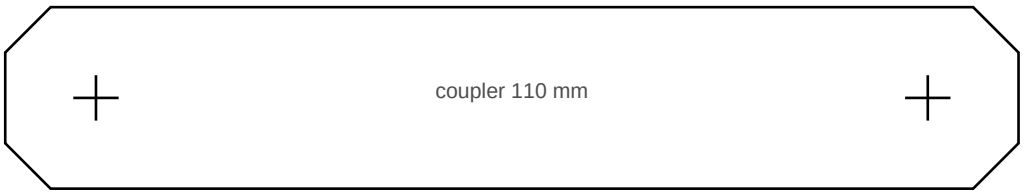
07 Four-bar linkage toy

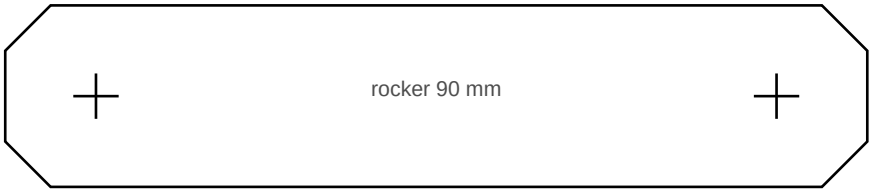
Fits on one A4 sheet. Cut the base and 3 links. Punch the + marks with a pencil or skewer. Join with split pins (paper fasteners): crank and rocker to the base, coupler on

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)





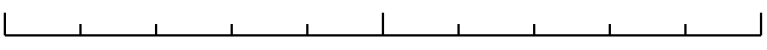




Cut the base and 3 links. Punch the + marks with a pencil or skewer.

Join with split pins (paper fasteners): crank and rocker to the base, coupler on top.

Turn the crank: it goes all the way round, the rocker swings back and forth.

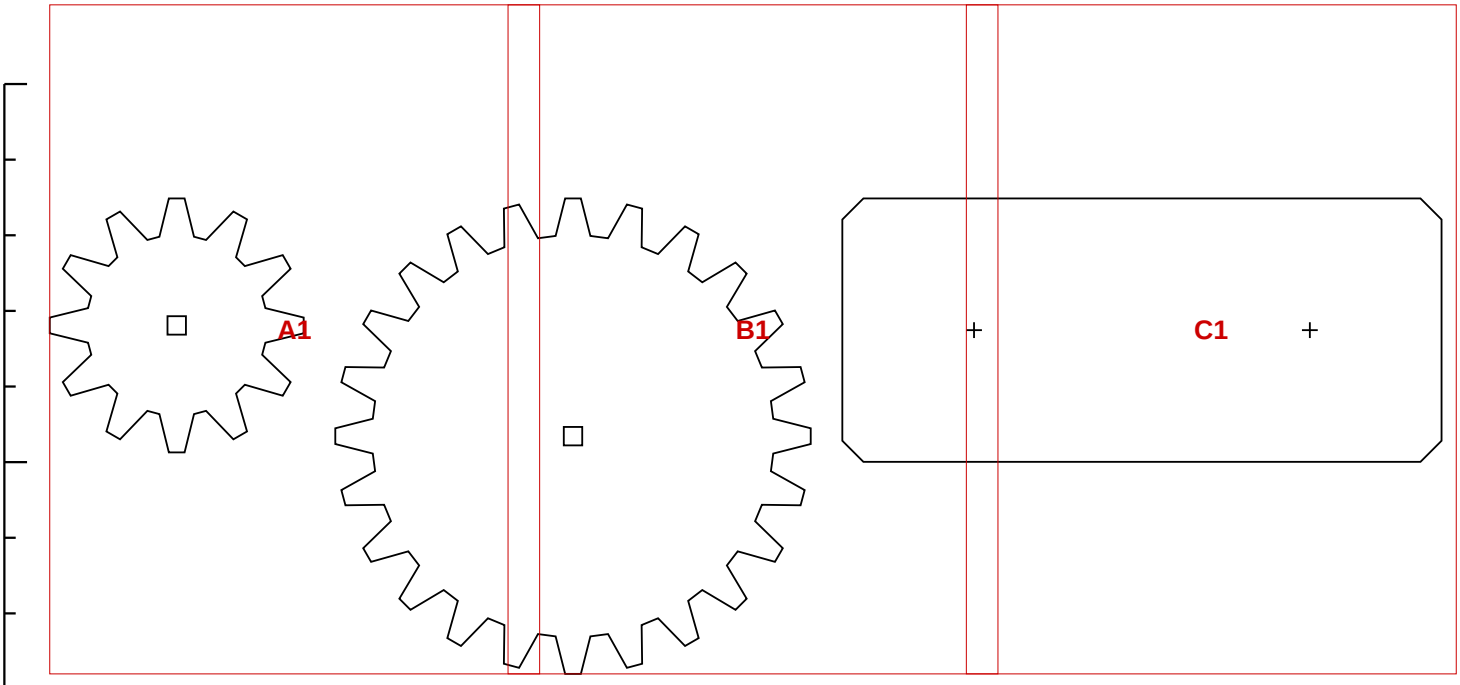


100 mm: measure it! (print at 100 %)

08 Gear pair 12T / 24T (2:1)

3 A4 sheets (3 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

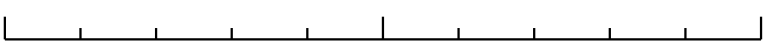
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: gear 12 teeth, gear 24 teeth, base, centers 127.5 mm apart

Cut both gears: each tooth is 4 straight cuts. Cut the 7 mm square center holes.

Push a pencil through each gear into the base marks (or use split pins).

Turn the small gear twice: the big gear turns once, the other way.

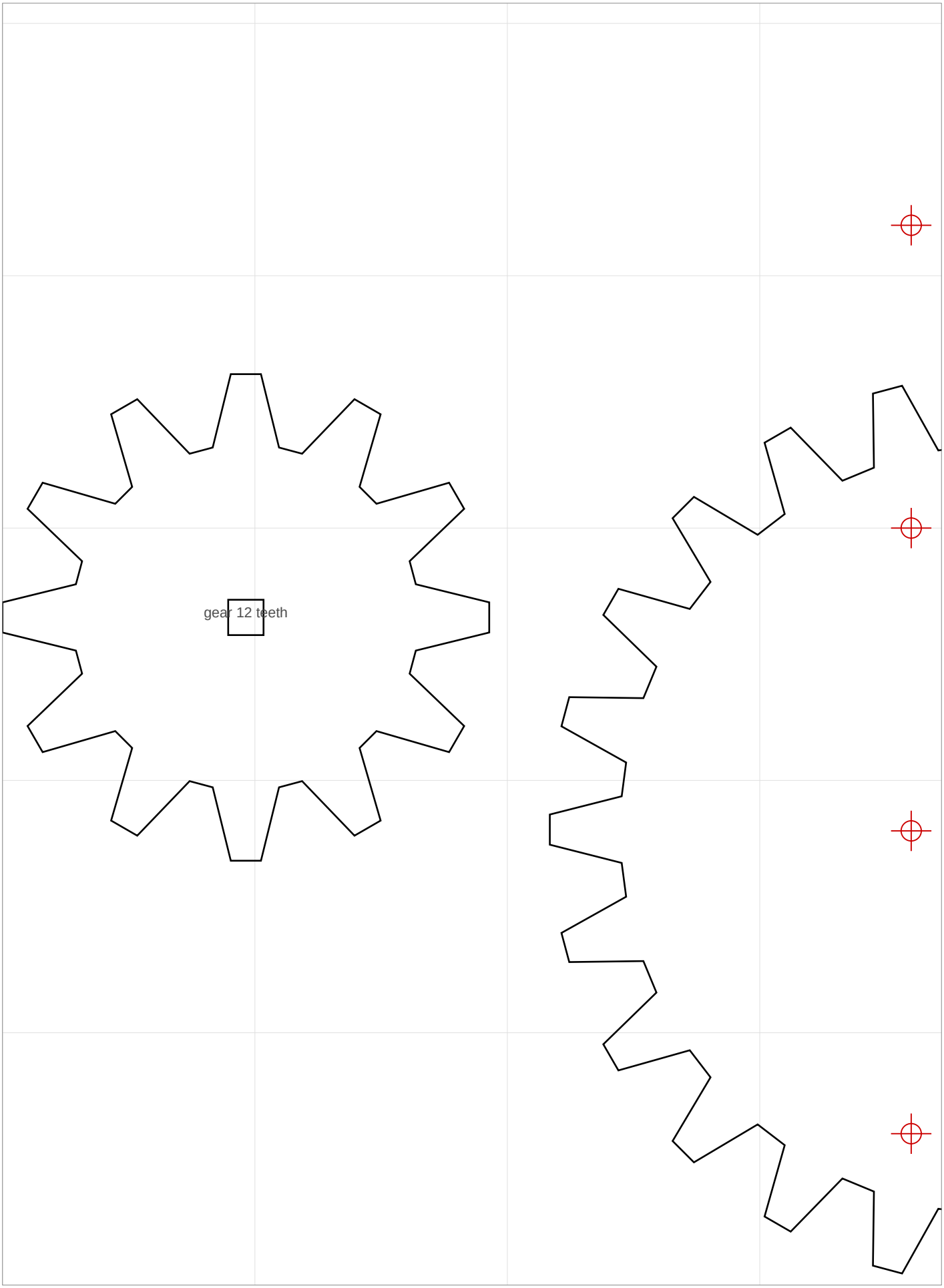


100 mm: measure it! (print at 100 %)

08 Gear pair 12T / 24T (2:1) - sheet A1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

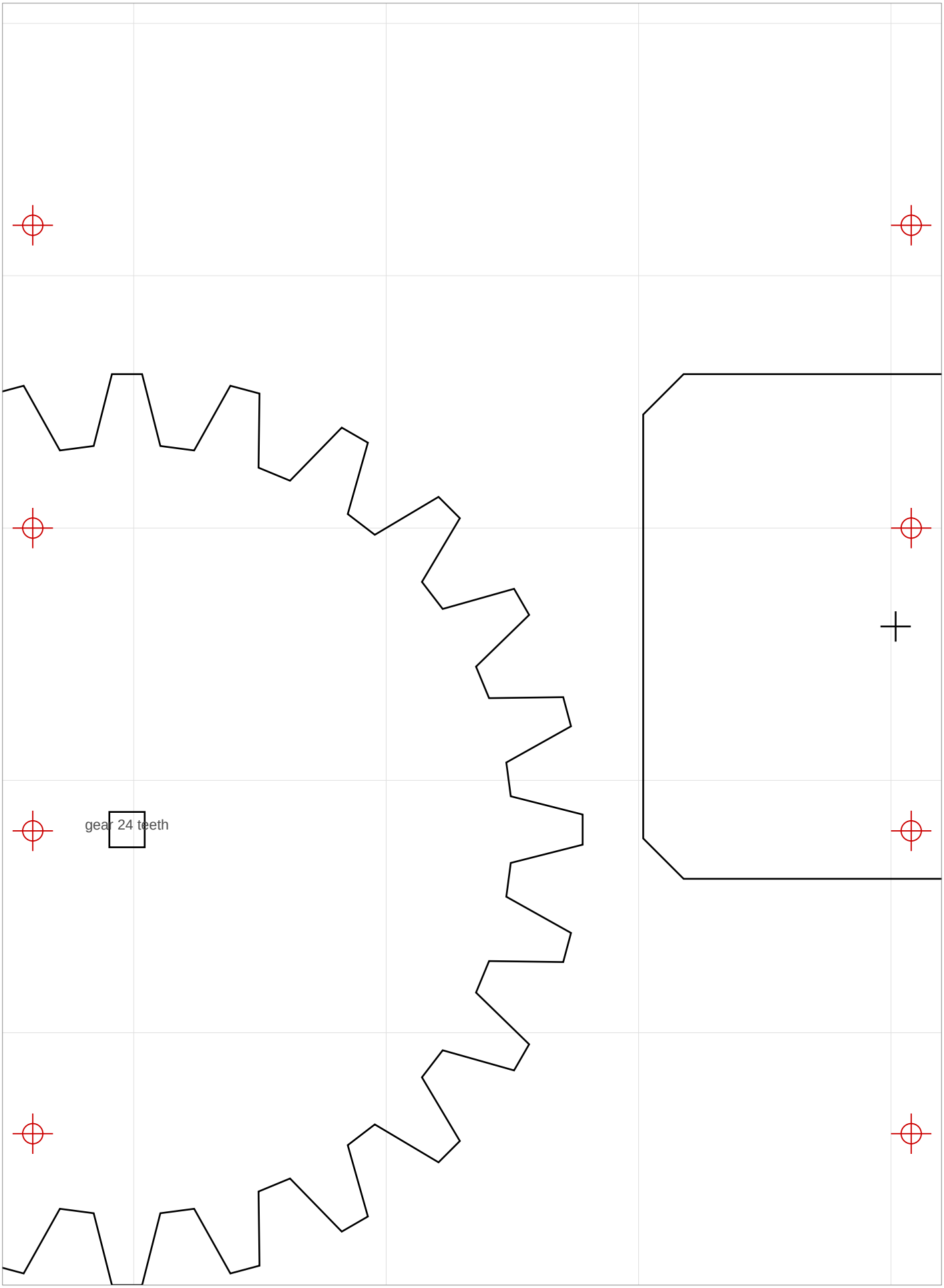


100 mm: measure it! (print at 100 %)

08 Gear pair 12T / 24T (2:1) - sheet B1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



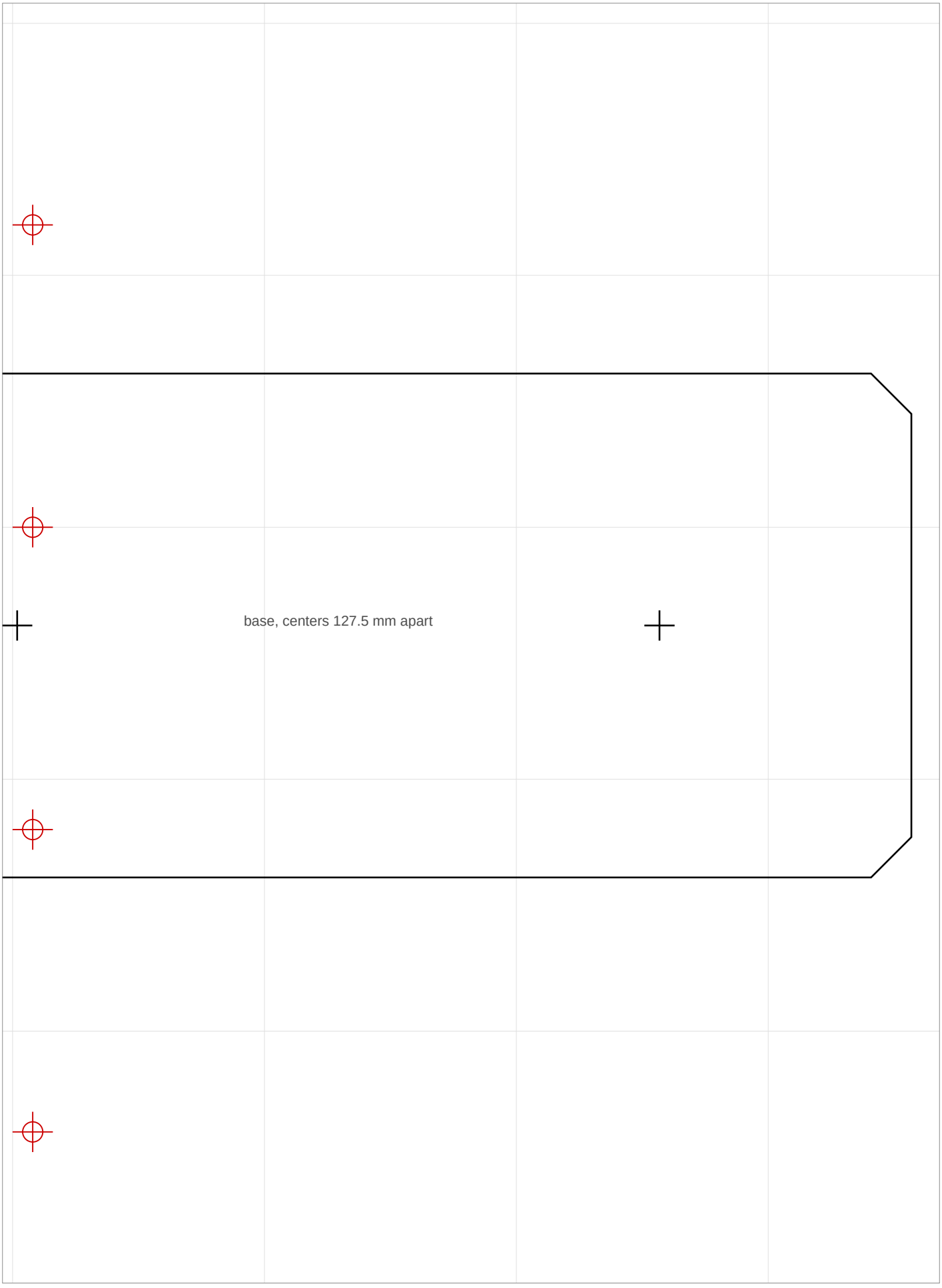
<- left edge overlaps sheet A1

100 mm: measure it! (print at 100 %)

08 Gear pair 12T / 24T (2:1) - sheet C1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



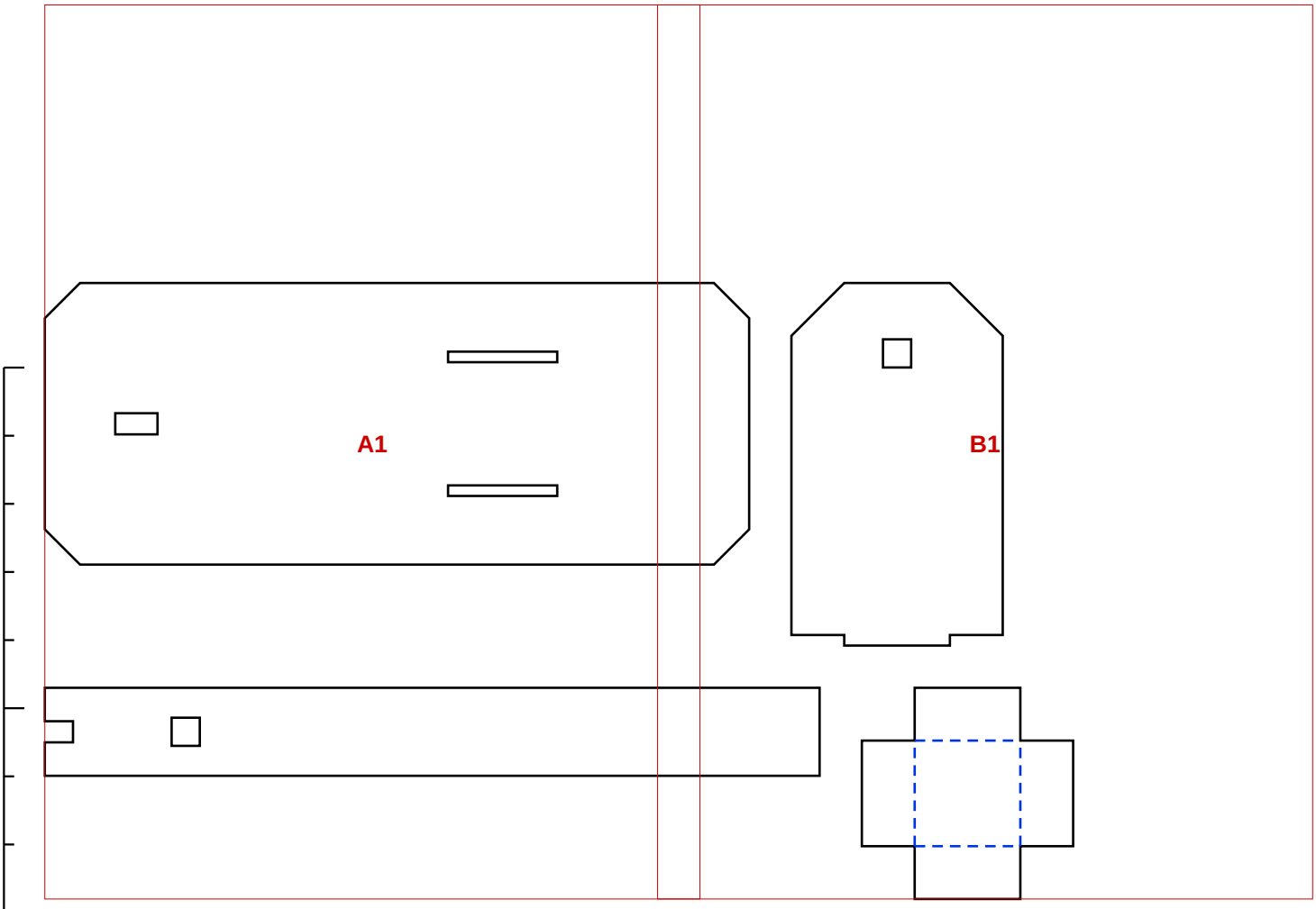
<- left edge overlaps sheet B1

100 mm: measure it! (print at 100 %)

09 Catapult

2 A4 sheets (2 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

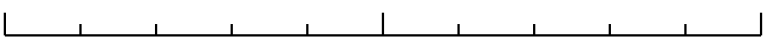
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: base, upright x2 (x2), arm, axle hole 40 mm from the back, cup: fold the 4 sides up, tape corners

Cut all parts; the base slots and square holes are cut out completely.

Push the upright tabs into the base slots (tape underneath). Pencil through uprights and arm.

Glue the cup to the front of the arm. Rubber band from the arm notch to the base hole.

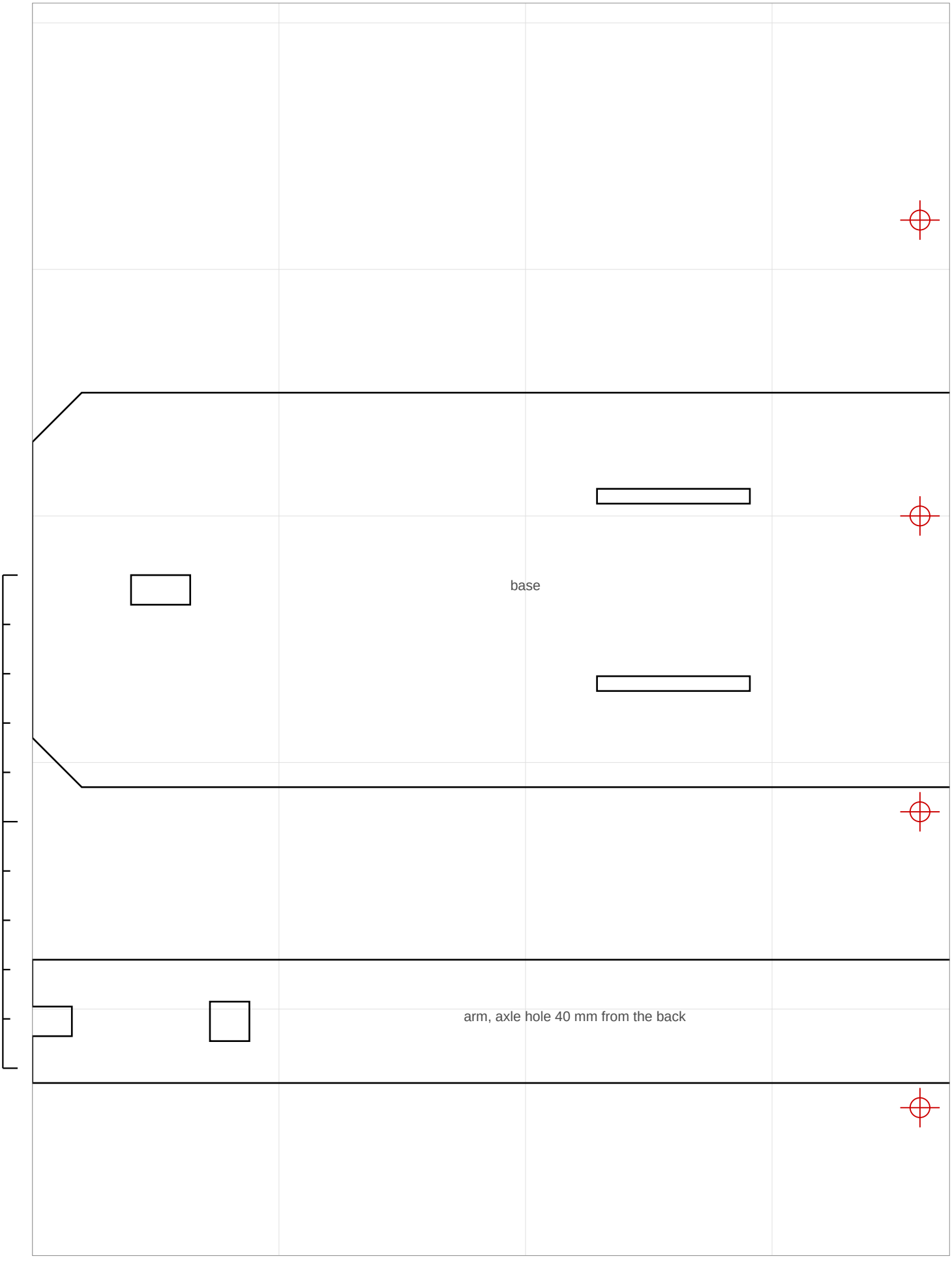


100 mm: measure it! (print at 100 %)

09 Catapult - sheet A1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

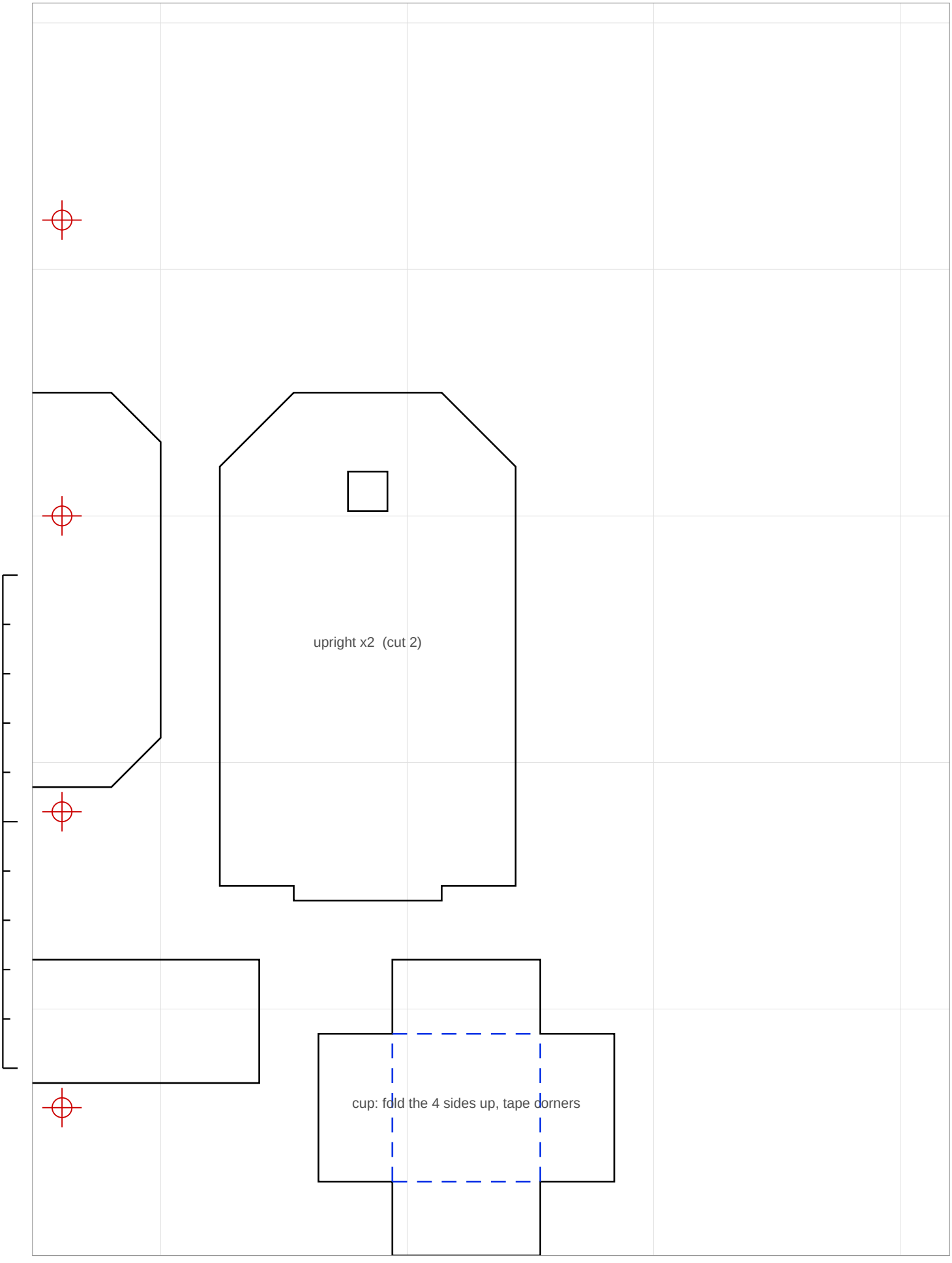


100 mm: measure it! (print at 100 %)

09 Catapult - sheet B1 (2 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



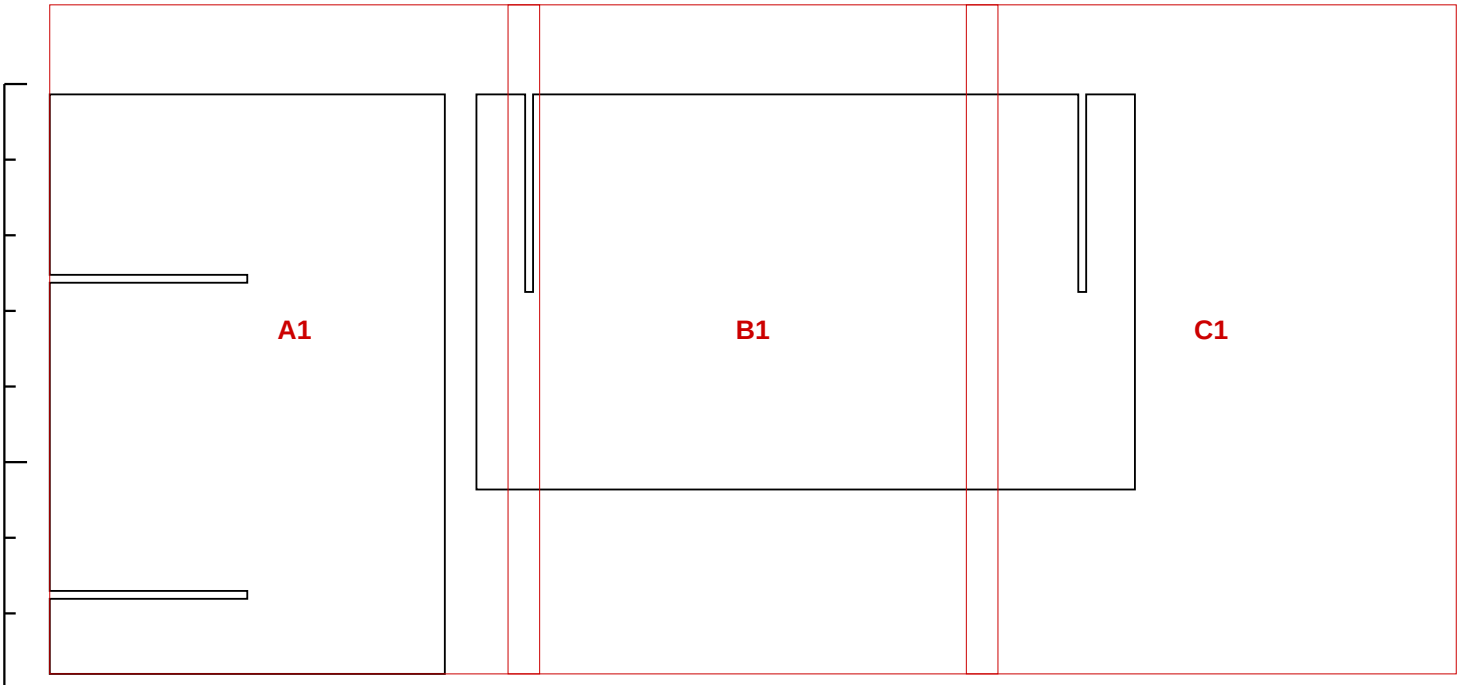
<- left edge overlaps sheet A1

100 mm: measure it! (print at 100 %)

10 Small shelf (250 x 150 x 220)

3 A4 sheets (3 across x 1 down) + this map. Print ALL pages at 100 % / actual size.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



How to join the sheets:

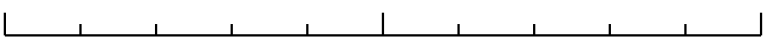
1. Cut each sheet along its gray frame on the LEFT and BOTTOM side only (not on A-column / bottom row).
2. Lay it over its neighbour so the red crosses sit exactly on top of each other. Tape it.
3. Join a whole row first, then the rows. Then glue the template to the board and cut.

Parts: side x2 (x2), shelf x2 (x2)

Cut 2 sides (flutes up-down) and 2 shelves (flutes left-right).

Slide each shelf's slots into the side slots from the front.

Light items only, or use double-wall board.

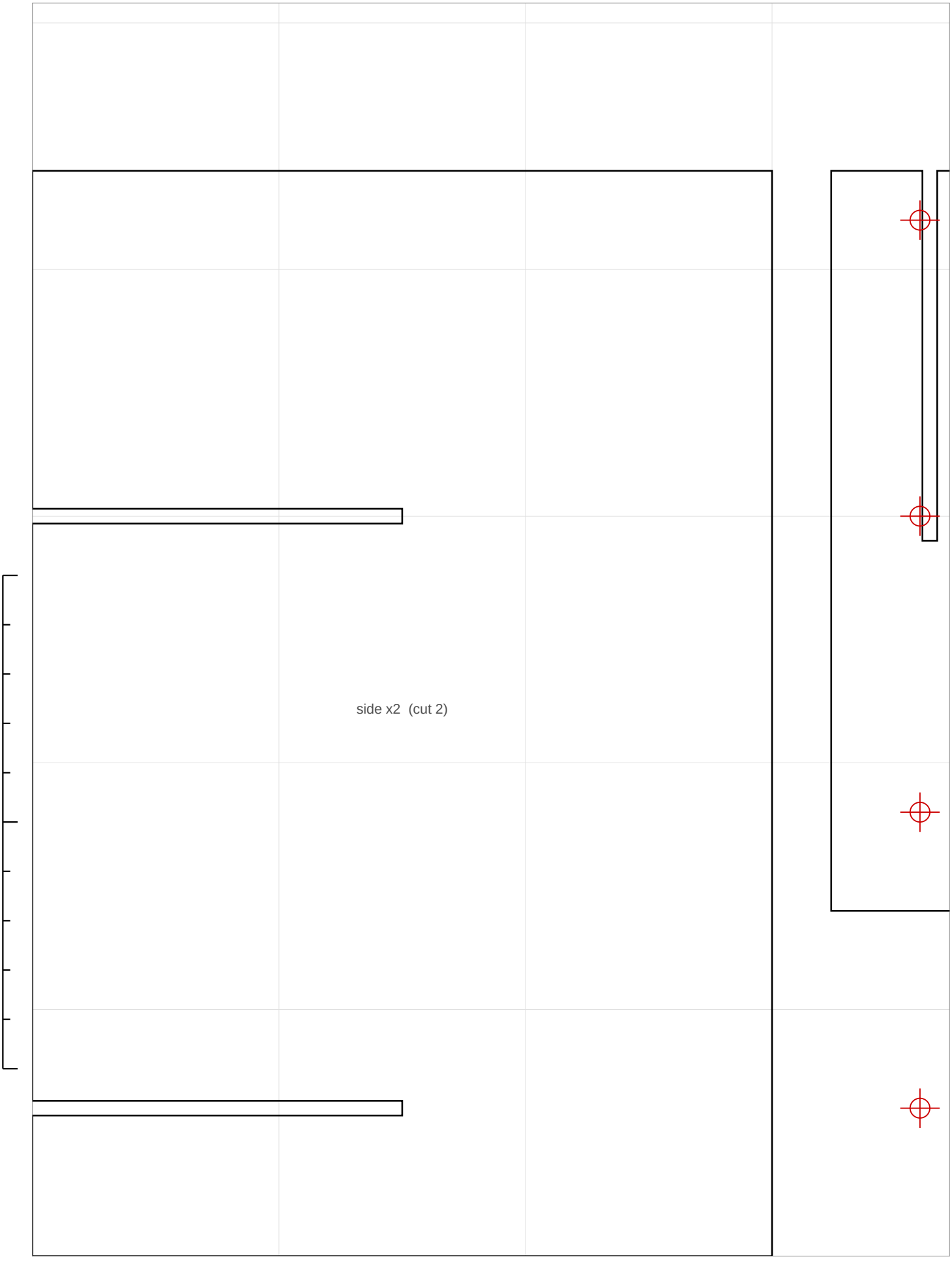


100 mm: measure it! (print at 100 %)

10 Small shelf (250 x 150 x 220) - sheet A1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)

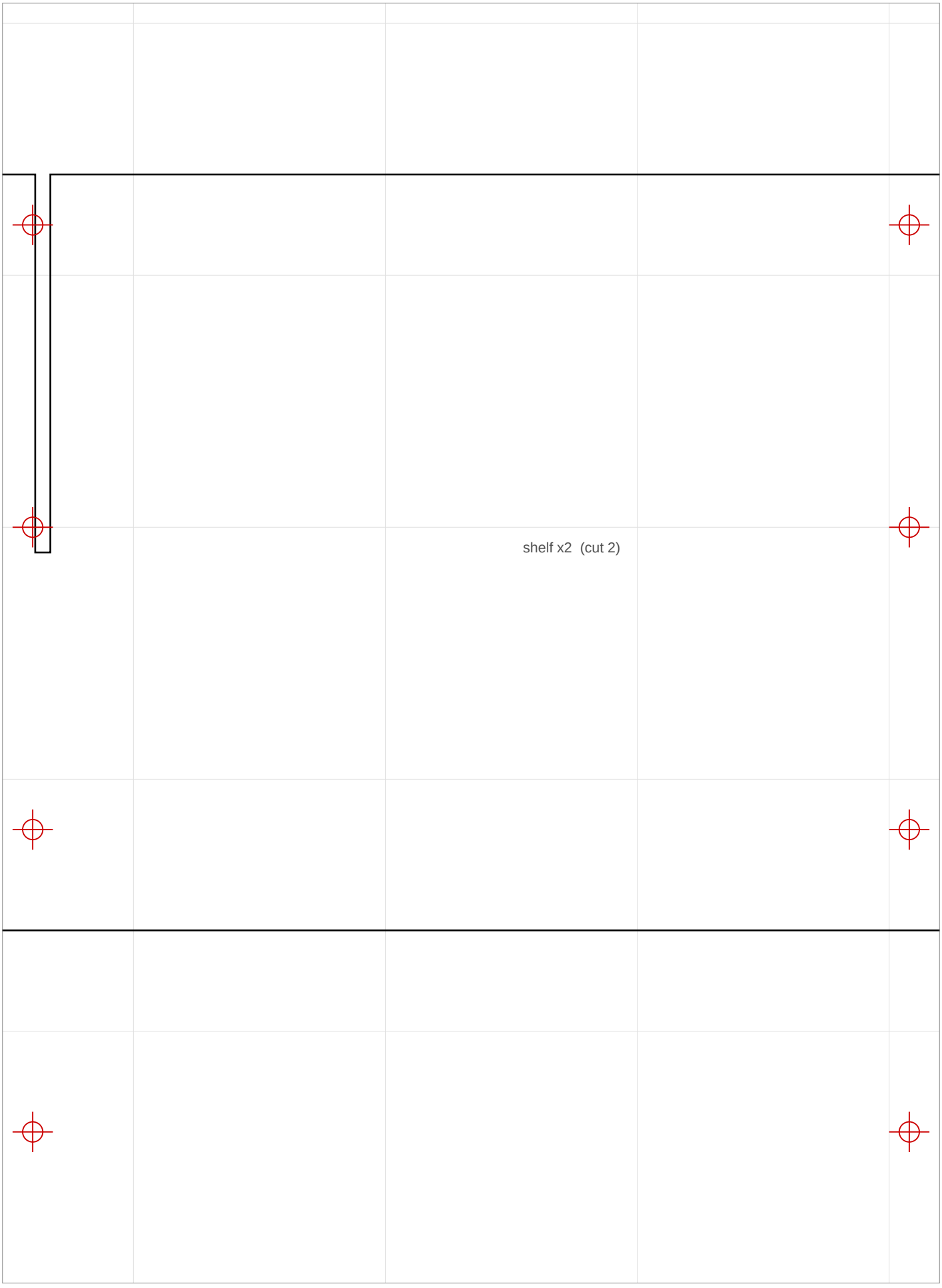


100 mm: measure it! (print at 100 %)

10 Small shelf (250 x 150 x 220) - sheet B1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



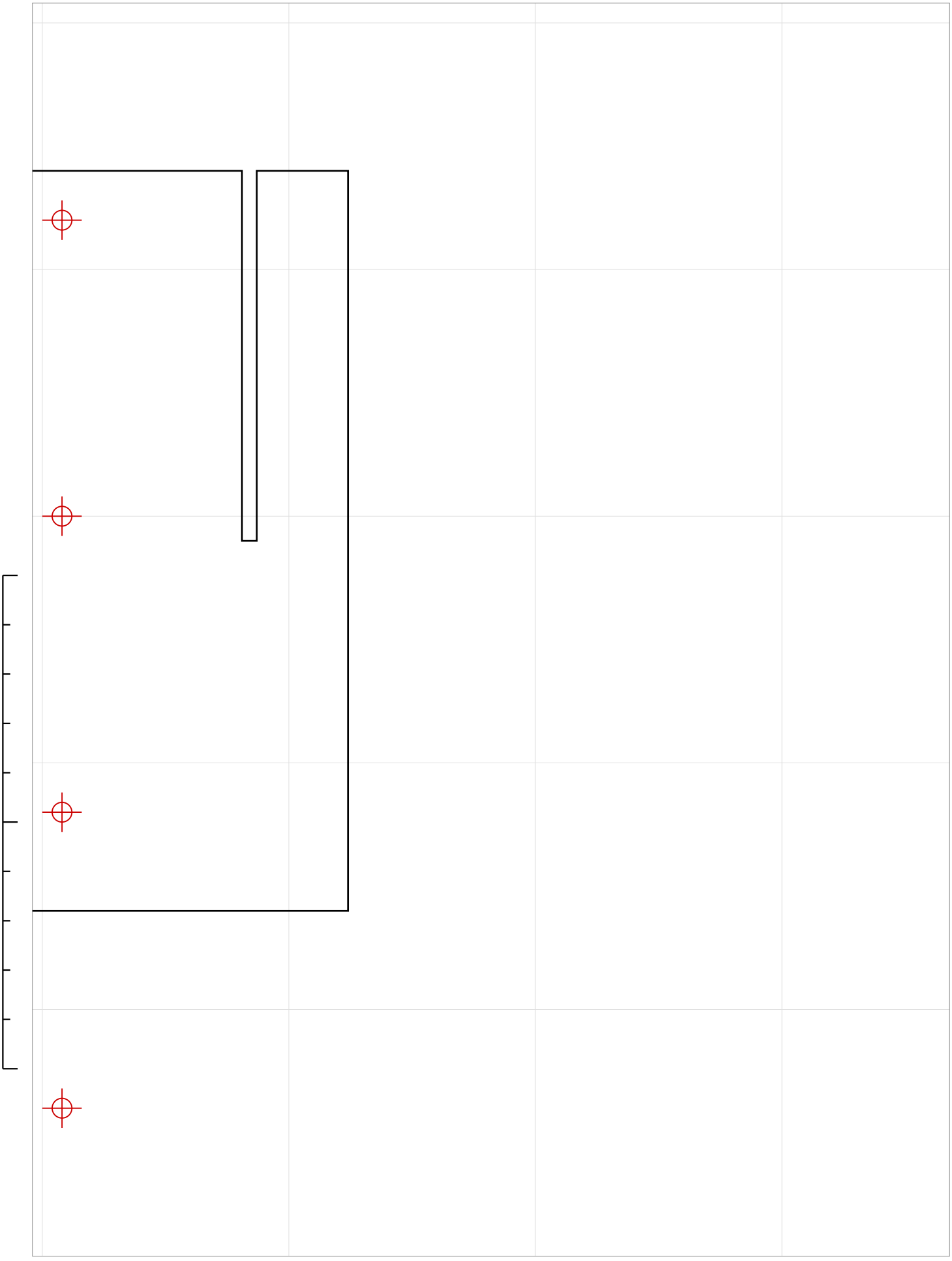
<- left edge overlaps sheet A1

100 mm: measure it! (print at 100 %)

10 Small shelf (250 x 150 x 220) - sheet C1 (3 x 1)

Print at 100 %. Trim the LEFT and BOTTOM edge along the gray frame, then overlap the neighbour and match the red crosses.

Solid = CUT dashed blue = SCORE + FOLD gray = GLUE + = punch hole Board 3 mm (slots 3 mm)



<- left edge overlaps sheet B1

100 mm: measure it! (print at 100 %)